

Groundwater pollution by Cystine manufacturing industrial effluent around the factory

S. Srinivasa Gowd · B. Kotaiah

Abstract The pollution potential of Cystine industrial effluents on groundwater pollution is assessed. The results indicate that the Cystine industrial effluent which is partially treated and stored in lagoons contains a high concentration of inorganic solids and soluble organics. The percolation of these effluents caused groundwater pollution impairing the water quality.

Key words Human hair · Cystine – manufacturing effluents · Groundwater pollution

Introduction

Srinivasa Cystine Industry produces Cystine, an amino acid from human hair as a basic raw material. The system of manufacturing consists of acid hydrolysis with HCl, precipitation with alkali, and purification with activated carbon and clays, washing, centrifugation etc., which is usually carried in an aqueous medium. The finished product is separated and the excess primary reagents and unwanted intermediate compounds are discharged into the factory drain. Frequent equipment and floor washings, and workers washing their hands further contribute wastewater generation in this factory. The wastewater contains a high concentration of acidity, volatile and suspended solids, BOD and COD, and sodium chloride. The generation of wastewater from all the sources is estimated about 35000 l/day (Niranjan and Kotaiah 1993).

The industry treats the wastewater partially by neutralization, coagulation and clarification followed by biological

treatment using an activated sludge process. Finally, the effluent is stored in unlined lagoons. The percolation of lagoon effluent into the ground has caused extensive groundwater pollution since 1984. There were many protests and demonstrations by the adjacent agriculturists for the closure of this manufacturer on the plea that their irrigation wells were contaminated by the factory effluent. Investigation of groundwater pollution around the factory by the authors provides the results presented in this paper.

Methodology of work

Preliminary investigations of the soil profile by a resistivity survey and of the groundwater flow movement by well inventory method were carried out (Karanth 1987). The distance of the wells to the factory site is established by measurement with a chain and compass survey. Samples of effluent were collected from the lagoons for analysis to assess the nature of contaminants percolating into the groundwater table. Groundwater samples from 20 dug wells around the factory were collected during pre-monsoon and post-monsoon periods (Fig. 1). All the samples were analyzed for chemical parameters according to the Standard Methods (APHA, AWWA, WPCF, 1995). Ca^{+2} , Na^{+} , and K^{+} were determined by flame photometer.

Results

The results of analyses of lagoon effluents are presented in Table 1 and indicate that the lagoon effluent contains a high concentration of inorganic and organic dissolved solids as well as suspended solids. Sodium chloride (NaCl) is the principle inorganic substance present, comprising about 99.85% of the total solids present in lagoon effluent. The COD/BOD ratio is about 15, indicating that the effluent is not easily amenable for biological degradation in a lagoon environment due to the presence of high sodium chloride content, which exert osmotic effect around the bacterial cell wall restricting the transport of nutrients into the cell for metabolism. The results of the average analysis of groundwater samples are presented in Tables 2 and 3 and indicate signifi-

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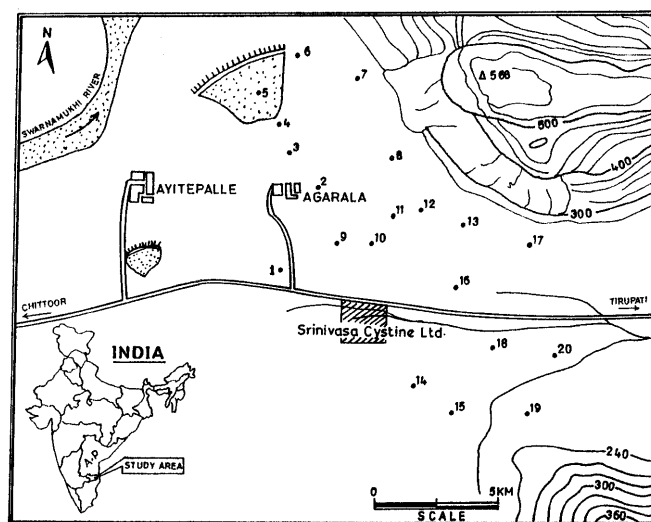
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Table 1

Characteristics of lagoon effluents

Sample	Parameter	Concentration
1	pH	7.0–7.4
2	Electrical Conductivity (micro mhos/cm)	230 400–265 000
3	Suspended solids	1200–2050
4	Inorganic dissolved solids	140 500–148 500
5	Total alkalinity (as CaCO ₃)	110–120
6	Total acidity (as CaCO ₃)	240–250
7	Total hardness (as CaCO ₃)	340–400
8	Sodium (Na ⁺)	55 000–58 300
9	Calcium (Ca ²⁺)	100–120
10	Magnesium (Mg ²⁺)	20–24
11	Chloride (Cl ⁻)	85 000–90 000
12	Sulphates (SO ₄ ²⁻)	50–60
13	COD	42 000–45 000
14	BOD ₅ at 20 °C	2800–3000

**Fig. 1**

Location map of the study area including sites at which water samples collected

cant variation of the quality of groundwater indicating the local groundwater pollution. Wells numbered as 1, 7, 19, and 20 are not affected by pollution with lagoon effluents, as they are not within the confluence of contaminant front. The contaminant front seems to flow in the direction of north-west and also to east. A comparison of the groundwater quality of the polluted and non-polluted waters indicates that the polluted water contains a high concentration of specific electrical conductance (dissolved solids), hardness, chlorides, and soluble COD. These

changes in the polluted groundwaters are possibly due to the contamination of Cystine industry effluents percolated into the groundwater table. The average infiltration factor of this area is around 12% which was estimated by Reddy and others (1993).

Geology

Geologically, the area is dominated by Archean Granites and basic intrusives such as Dolerite dykes (King 1872; Rama Mohana and others 1987). Granites are coarse- to

Table 2

Chemical constituents in groundwater samples (pre-monsoon)

Well	pH	SEC	IDS	Total alkalinity as CaCO ₃	Total acidity as CaCO ₃	Total hardness as CaCO ₃	Na ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	SO ₄ ²⁻
1	7.4	650	360	174	12	150	30	42	11	46	20
2	7.5	2800	1570	334	24	190	420	65	7	646	30
3	7.3	2530	1440	387	20	210	340	63	13	524	40
4	8.2	1780	980	347	12	290	175	38	47	272	30
5	7.7	1740	970	403	0	180	160	21	31	250	30
6	7.3	1960	1110	449	8	120	200	19	18	320	20
7	7.1	930	540	298	8	150	46	51	6	70	10
8	7.2	3560	1960	243	16	1200	500	214	162	772	20
9	6.1	9230	5070	426	48	2800	1340	927	118	2160	20
10	7.9	9940	5560	184	12	1530	1900	289	196	2950	10
11	7.7	9050	4980	230	20	2100	1580	392	272	2440	20
12	7.8	3550	1990	236	12	1000	550	180	140	850	20
13	7.8	4380	2530	279	16	1450	700	265	195	1075	20
14	6.6	5630	3150	295	24	1900	850	406	215	1310	20
15	7.1	5610	3080	328	28	1650	820	490	99	1265	10
16	7.5	2960	1630	279	20	1120	340	185	160	536	30
17	7.5	5460	3060	285	28	1680	855	291	232	1320	20
18	7.8	1430	780	243	12	830	90	150	110	140	30
19	8.0	780	430	233	8	190	25	48	17	42	20
20	7.5	1490	830	315	20	340	135	35	10	210	20

Table 3

Chemical constituents in groundwater samples (post-monsoon)

Well	pH	SEC	IDS	Total alkalinity as CaCO ₃	Total acidity as CaCO ₃	Total hardness as CaCO ₃	Na ⁺	Ca ⁺²	Mg ⁺²	Cl ⁻	SO ₄ ⁻²	COD
1	6.5	810	446	167	20	188	60	69	4	90	20	392
2	6.0	3110	1775	276	46	1070	430	103	109	700	10	304
3	6.0	2930	1700	266	56	980	420	290	65	669	20	244
4	6.7	1740	1640	325	36	540	190	110	65	289	30	0
5	6.1	1740	970	344	12	570	155	180	25	240	30	148
6	6.3	1980	1130	344	12	540	200	115	60	315	60	28
7	6.1	790	430	197	4	250	45	67	22	59	5	24
8	6.4	3820	2140	128	14	1260	620	253	142	961	10	12
9	6.1	12825	7050	262	40	2968	2180	935	155	3365	10	296
10	6.7	5050	2820	138	10	1696	835	310	225	1290	5	144
11	5.6	7480	4190	220	22	2480	1230	470	310	1897	10	328
12	5.6	6820	3880	272	22	1130	1260	250	178	1957	5	188
13	5.6	6950	4030	256	28	1340	1300	285	100	2015	10	328
14	6.2	7800	4290	262	42	2700	1220	530	335	1890	5	340
15	6.1	7905	4500	279	62	2640	1280	670	235	1980	5	332
16	6.4	4520	2670	236	20	1060	800	225	110	1220	20	200
17	6.2	5040	2820	200	32	1790	770	390	195	1190	20	280
18	6.7	1160	690	213	18	460	100	40	85	165	20	272
19	6.8	930	520	184	6	320	70	60	45	100	20	316
20	6.4	1430	820	276	26	320	150	40	12	220	20	324

medium-grained and porphyritic type of texture is noticed with feldspars as phenocrysts. The essential mineral constituents present in granites are quartz (SiO₂) and feldspar K, Na, Ca (AlSi₃O₈), with Hornblende (Ca, Mg, Fe, Na, Al)₇₋₈ (Al, Si)₈ O₂₂, Mica K(Mg, Fe)₃ (Al, Si)₃ O₁₀ (OH, F)₂, and Augite (Ca, Mg, Fe, Al)₂ (Al, Si)₂ O₆, as major accessories. Basic intrusives such as Dolerite dykes are traversing through the granites. The mineral composition of dyke rock is Augite, Plagioclase feldspars (Na, Al, Si₃O₈ – Ca, Al₂, Si₂ O₈) and ferro-magnesium minerals (Garrels 1967; Deer and others 1979).

The vertical profile of the unsaturated zone consists of various distinct layers (Table 4). The soil profile consists of:

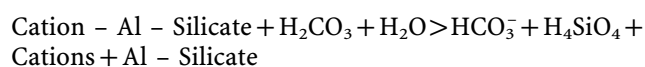
1. A-horizon extends up to a depth of 35 cm and is composed of yellowish-brown to light brown, fine, sandy loam. It is loose and moist with good root penetration.
2. B-horizon extends from 35 to 120 cm depth and is composed of reddish-brown coarse sandy loam with low percentage of clay. It is fairly compact with blocky

structure with good porosity and crumbles under pressure. Root penetration is good.

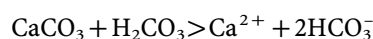
3. C-horizon extends from 120 to 180 cm depth and is interpreted by reddish brown to dark yellowish brown colour. It is massive, plastic, moist and has fairly good drainage (Reddy 1982).

Occurrence of groundwater

During the percolation of effluents into the ground, degradable organics are decomposed producing carbon dioxide and other organic acids. The carbon dioxide thus produced would form acid water (Carbonic acid, H₂CO₃), which is expected to attack the carbonate and other weathered minerals of the Earth's mantle. This causes a release of cations (Na⁺, K⁺, Ca²⁺, Mg²⁺ etc.) and bicarbonate ion as dissolved solids into the percolated water. This attack can be represented in a general form as:



and



Occurrence and movement of pollutants

The inorganic ions such as sodium, potassium, chloride, sulphates, bicarbonate and water soluble organics (COD) present in the effluents are not attenuated by the Earth's minerals and passed directly into the groundwater table for dilution. This would lead to subsequent high mineral content in the native groundwaters which has been clearly reflected in the analysis of well waters at polluted zones.

Table 4

Vertical profile of the unsaturated zone

Sample	Thickness of layer (m)	Nature of formation
1	0–2	Sandy loam
2	2–10	Semi-weathered, fractured granite
3	10–35	Weathered granite
4	>35	Hard rock

Conclusion

1. The lagoon effluent contains a high concentration of inorganic solids and soluble organics. Sodium chloride is not attenuated during the percolation of effluent through the Earth's mantle and directly leaches into the groundwater table.
2. The carbonic acid produced during the percolation of the effluents attacks the Earth's mantle and extracts the mineral ions into the solution. This causes leaching of Ca^{+2} , Mg^{+2} , Na^{+} , Fe^{+2} , Al^{+3} , F^{-} , and HCO_3^{-} ions causing high hardness and dissolved solids in the native groundwater.
3. The wells located in the northwest and east direction are most effected by pollution of the lagoon effluents.
4. Some of the wells numbered as 1,7,19, and 20 are not affected by pollution indicating that these wells are not with in the confluence of the contaminant front.

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